

Supplementary information

Transferable 3D Convolutional Neural Networks for Elastic Constants Prediction in Nanoporous Metals

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1 Generation of nanoporous structures

Atomistic 3D nanoporous configurations were generated by implementing the algorithm proposed by Soyarslan et al. [1], which involves superposition of standing sinusoidal waves of fixed wavelength, random phase, and specific directions. The corresponding random field generator is mentioned in Eq. S1

$$f(\vec{x}) = \sqrt{\frac{2}{N}} \sum_{i=1}^N \cos(\vec{q}_i \cdot \vec{x} + \phi_i) \quad (\text{S1})$$

where $\vec{x}$ is the position vector, $\vec{q}_i$ and ϕ_i denote the wave direction and phase, respectively and N indicates the number of waves involved in the generation of the nanoporous microstructure. A specific set of vectors $\vec{q}_i = \frac{2\pi}{a}(h, k, l)$ was defined based on a constant value $H = \sqrt{h^2 + k^2 + l^2}$, where h, k, l are integers corresponding to the Miller indices and a is the edge length of the simulation domain, and therefore $|\vec{q}| = q_0 = \frac{2\pi H}{a}$.

In this study, $H^2 = 9$, the number of repetitions of the unit cell = 125, and the domain size is given by $a = \text{lattice constant} \times 125$. Accordingly, the 3D periodic nanoporous structures were generated using the random field defined in Eq. S1. For $H^2 = 9$, there are 30 wave directions and they are mentioned in Table S1.

Table S1: Wave direction vector list corresponding to $H^2 = 9$

Index	h	k	l
1	-3	0	0
2	-2	-2	-1
3	-2	-2	1
4	-2	-1	-2
5	-2	-1	2
6	-2	1	-2
7	-2	1	2
8	-2	2	-1
9	-2	2	1
10	-1	-2	-2
11	-1	-2	2
12	-1	2	-2
13	-1	2	2
14	0	-3	0
15	0	0	-3
16	0	0	3
17	0	3	0
18	1	-2	-2
19	1	-2	2
20	1	2	-2
21	1	2	2
22	2	-2	-1
23	2	-2	1
24	2	-1	-2
25	2	-1	2
26	2	1	-2
27	2	1	2
28	2	2	-1
29	2	2	1
30	3	0	0

2 Extraction of morphological and topological descriptors

The morphological and topological descriptors of the nanoporous structure were extracted from the skeletonized network, which was obtained through post-processing of the initially generated 3D nanoporous models. First, a 3D image of the nanoporous structure was created using voxels, with each voxel sized approximately equal to the

diameter of an atom. The voids between the voxels were filled to ensure a continuous and well-defined representation of the porous network. A 3D skeletonization algorithm [2] was then applied to this voxel-based model, reducing it into a skeleton framework while preserving the topology of the porous network. Finally, the skeletonized structure was converted back into coordinate data, enabling further analysis and visualization. The descriptors were then derived by integrating the skeleton elements and nodes with the corresponding nanoporous volume. An example of a fully periodic nanoporous structure from atomistic simulations is presented in Fig. S1a. The corresponding skeleton, segmented into discrete points, is shown in S1b, where points located at node centers are highlighted in red.

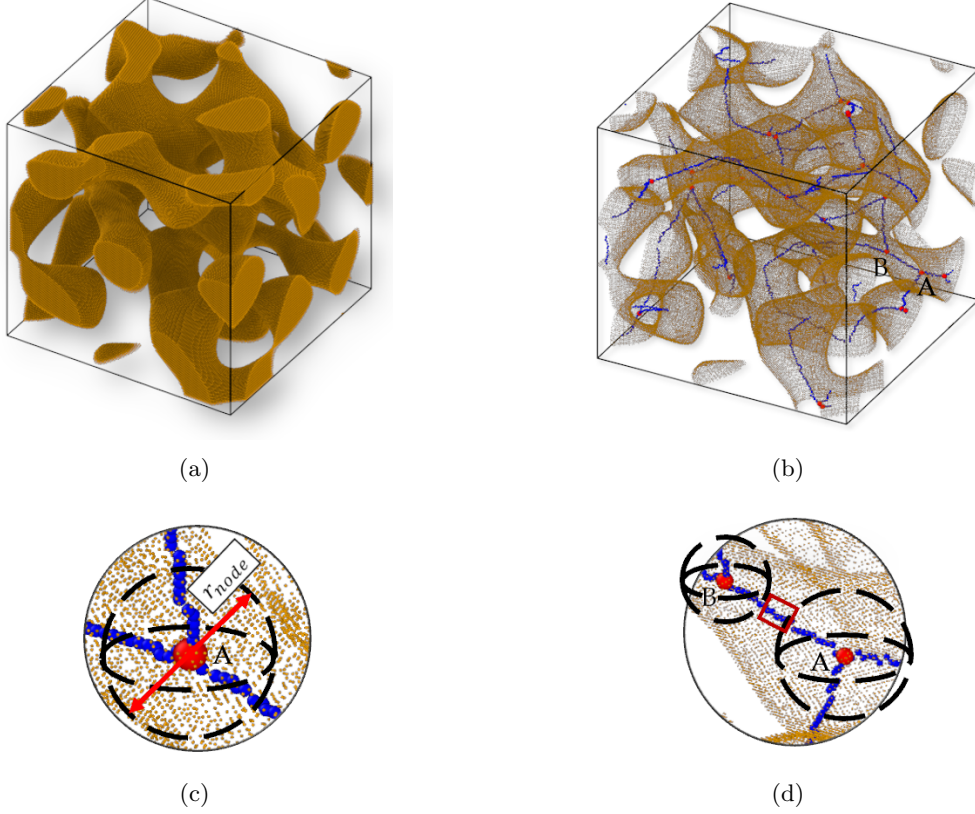


Fig. S1: Topological descriptors. (a) An atomic nanoporous structure. (b) The skeleton, superimposed with the surface atoms of the nanoporous structure. (c) The sphere defines the size of the node at point A (marked in (b)). (d) Two adjacent nodes (A and B, marked in (b)) and the skeleton elements are considered as the ligaments in between the nodes.

Leveraging the skeleton, several directional descriptors are evaluated in this study. The first among them are the ligament-only directional resistance (LDR) and its modified form (MLDR), both defined in the main text, similar to one defined in Ref. [3]. These descriptors assume that all skeleton elements, connecting pairs of skeleton points, constitute ligaments, thereby neglecting the finite size of the nodes. The descriptors are defined as

$$\begin{aligned} LDR &= \frac{\sum_{\text{ligaments}} l_i \cos \gamma_i}{L}, \\ MLDR &= \frac{\sum_{\text{ligaments}} l_i \cos^{10} \gamma_i}{L}, \end{aligned} \quad (\text{S2})$$

where l_i is the length of the skeleton elements, γ_i is the angle between the skeleton element and the loading axis, and L is the size of simulation box along the loading axis.

The previous approach implicitly assumes that the mechanical response of skeleton elements is uniform across ligament centers and node centers, while nodes contributing an isotropic response to deformation, as opposed to ligaments. To account for this, we reclassified skeleton elements, distinguishing those corresponding to ligaments from those residing within nodes. Specifically, a node is defined as the largest sphere centered on a skeleton node and fully contained within the nanoporous volume. An illustrative example is provided in Fig. S1c, for the node labeled A in Fig. S1b. Skeleton points lying outside these spheres are identified as ligaments, such as those within the red rectangle in Fig. S1d, corresponding to the ligament connecting nodes A and B. Notably, in the current study, under this definition, ligaments are substantially shorter than in the previous approach, and their lengths are less than the node size r_{node} . Based on this redefinition, and recognizing that nodes respond isotropically to mechanical loading, we introduce two new directional descriptors: the directional resistance (DR) and the modified directional resistance (MDR),

$$\begin{aligned} DR &= \frac{\sum_{\text{ligaments}} l_i \cos \gamma_i + \sum_{\text{nodes}} 2r_{\text{node},j}}{L}, \\ MDR &= \frac{\sum_{\text{ligaments}} l_i \cos^{10} \gamma_i + \sum_{\text{nodes}} 2r_{\text{node},j}}{L}. \end{aligned} \quad (\text{S3})$$

The summation over ligaments includes only those skeleton elements that lie outside the node volumes, while the summation over nodes includes the full set of skeleton nodes.

In the above definition, ligament contributions are considered independent of their cross-sectional area, implying that thick and thin ligaments contribute equally if oriented at the same angle relative to the loading axis. To incorporate geometric variability, we calculate the mean cross-sectional radius r_i , defined perpendicular to each skeleton element classified as a ligament. To capture both length and cross-sectional area, we introduce modified descriptors analogous to Eqs. S2 and S3, replacing length-based terms with volumetric ones. Accordingly, two new descriptors are defined: ligament-only volumetric resistance (LVR) and modified ligament-only volumetric

resistance (MLVR), both excluding contributions from node regions,

$$\begin{aligned} LVR &= \frac{\sum_{ligaments} \pi l_i r_i^2 \cos \gamma_i}{V}, \\ MLVR &= \frac{\sum_{ligaments} \pi l_i r_i^2 \cos^{10} \gamma_i}{V}. \end{aligned} \quad (S4)$$

Here, V denotes the total volume of the simulation box. Similarly, when the volumes of the nodes are taken into account, we define two additional descriptors: volumetric resistance (VR) and modified volumetric resistance (MVR). These descriptors incorporate both ligament and node contributions, providing a more comprehensive measure of directional mechanical resistance that accounts for topology and the cross-section area of the ligaments.

$$\begin{aligned} DR &= \frac{\sum_{ligaments} \pi l_i r_i^2 \cos \gamma_i + \sum_{nodes} \frac{4}{3} \pi r_{node,j}^3}{V}, \\ MDR &= \frac{\sum_{ligaments} \pi l_i r_i^2 \cos^{10} \gamma_i + \sum_{nodes} \frac{4}{3} \pi r_{node,j}^3}{V}. \end{aligned} \quad (S5)$$

3 Best-performing architecture (DenseNet-201)

The best-performing model in this study is a 3D version of the DenseNet-201 architecture. The network uses an initial convolutional layer with 64 filters followed by four dense blocks with 6, 12, 48, and 32 dense layers, and a growth rate of 32. Dense layers use a bottleneck convolution with a $1 \times 1 \times 1$ kernel followed by a $3 \times 3 \times 3$ convolution. Transition layers consist of a $1 \times 1 \times 1$ convolution and $2 \times 2 \times 2$ average pooling. The model maps a single-channel voxelized nanoporous structure to a single scalar output (C_{ii}). A scatter plot of MD-computed and predicted values of elastic constants (C_{11} , C_{22} , and C_{33}) is shown in Figure S2 together with a histogram of prediction errors.

Model training was performed with MSE as the loss function. The training loss is computed on the training set, and after each epoch the model is evaluated on the validation set to compute the validation loss. We monitor validation loss to select the best checkpoint: training continues while the validation loss decreases, but it is not stopped at the first validation increase. Instead, we allow a patience of 10 epochs - if the validation loss does not improve below the current best value within that period, training is terminated and the checkpoint corresponding to the lowest observed validation loss is restored. The per-epoch train, test, and validation loss demonstrating stable convergence and the absence of overfitting are shown in Figure S3.

DenseNet blocks with dense connectivities promote feature reuse throughout the whole network and ensure gradient flow through deep 3D stacks, enabling training of the model without vanishing gradients. The growth rate determines how many new filters (channels) each layer contributes, while bottleneck convolutions compress intermediate channels to reduce computation and memory costs without sacrificing

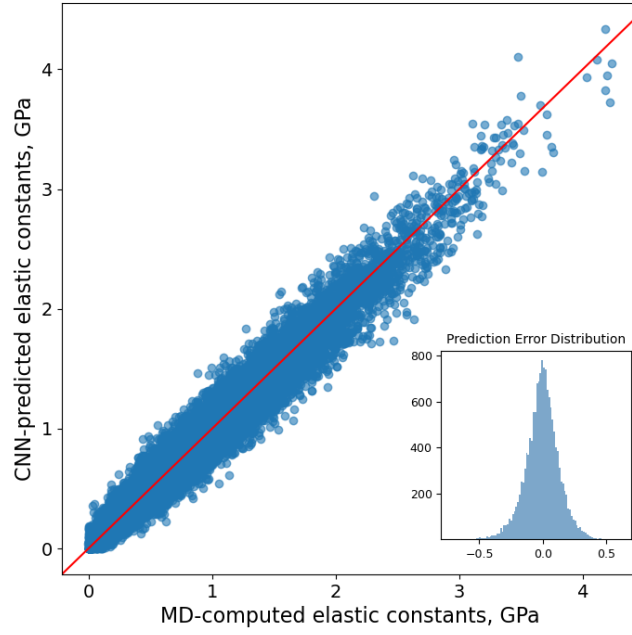


Fig. S2: MD-computed vs CNN-predicted elastic constants for 8 cross-validation folds with distribution of errors.

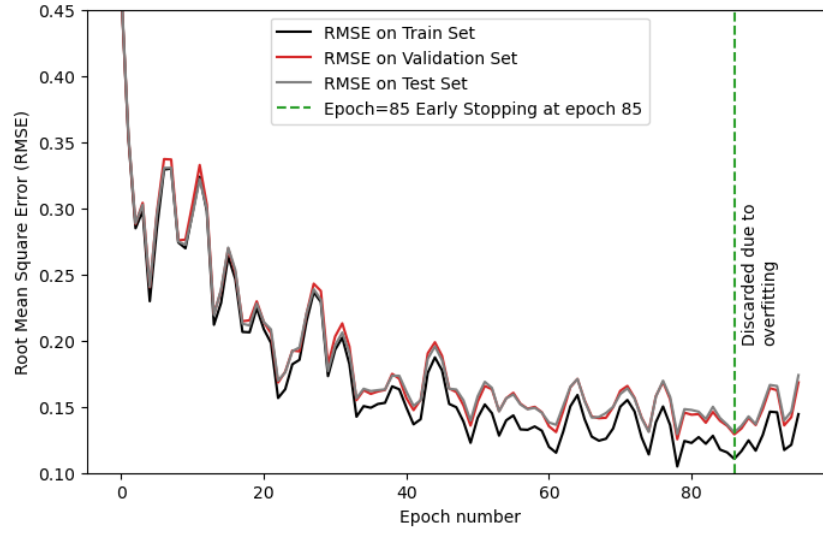


Fig. S3: Train, test, and validation loss during the training on Dataset 1.

predictive power. Convolutions with the kernel size of $3 \times 3 \times 3$ were used to extract local morphological features. Stacking these small kernels with downsampling increases the receptive field of the model so that deeper layers capture mesoscale and near-global patterns. Transition layers ($1 \times 1 \times 1$ convolution followed by average pooling) reduce channels and downsample data resolution. Batch normalization is used to stabilize training in the deep 3D network, and a final global average pooling reduces spatial dimensions to a small fully connected regressor that outputs a single value of elastic constant. Optional dropout in the dense layers is used for additional regularization and prevention of overfitting. Layer-by-layer data shapes are provided in Table 2.

Table 2: Layer by layer output shapes and channels of DenseNet-201 for an input size of 80^3 .

Layer	Output shape	Channels
Input	$80 \times 80 \times 80$	1
Conv1 ($7 \times 7 \times 7$, stride=(1,2,2))	$80 \times 40 \times 40$	64
MaxPool ($3 \times 3 \times 3$, stride=2)	$40 \times 20 \times 20$	64
DenseBlock1 (6 layers)	$40 \times 20 \times 20$	256
Transition1 (1×1 conv + AvgPool)	$20 \times 10 \times 10$	128
DenseBlock2 (12 layers)	$20 \times 10 \times 10$	512
Transition2	$10 \times 5 \times 5$	256
DenseBlock3 (48 layers)	$10 \times 5 \times 5$	1792
Transition3	$5 \times 2 \times 2$	896
DenseBlock4 (32 layers)	$5 \times 2 \times 2$	1920
BatchNorm + ReLU	$5 \times 2 \times 2$	1920
Global AvgPool	$1 \times 1 \times 1$	1920
Fully Connected	1×1	1 (elastic constant)

3.1 Model interpretability

In order to understand which parts of the NPG structure have the highest influence on the final prediction, a SHAP analysis of the DenseNet-201 model was performed. SHAP is a feature attribution method based on cooperative game theory, which assigns each input feature a contribution score reflecting its impact on the model’s output [4]. For high-dimensional models such as 3D CNNs, exact computation of Shapley values is infeasible, and SHAP employs approximation strategies (e.g., DeepSHAP, [4, 5]) that propagate attributions through the network layers. In our case, the CNN input is a voxelized representation of the structure, and SHAP values measure the marginal contribution of each structural element to the predicted stiffness. Positive values indicate features that increase the predicted modulus, while negative values correspond to features that decrease it. By aggregating these values across samples, we can identify which structural motifs consistently drive the model’s prediction and thereby possibly gain physical insight.

Figure S4 illustrates three orthogonal intersection planes through the same structure, taken perpendicular to the direction of the applied stress. In the top row, the binary structure is shown as a 2D cross-section, while in the bottom row the corresponding voxel-resolved SHAP values are plotted. Regions with positive SHAP values,

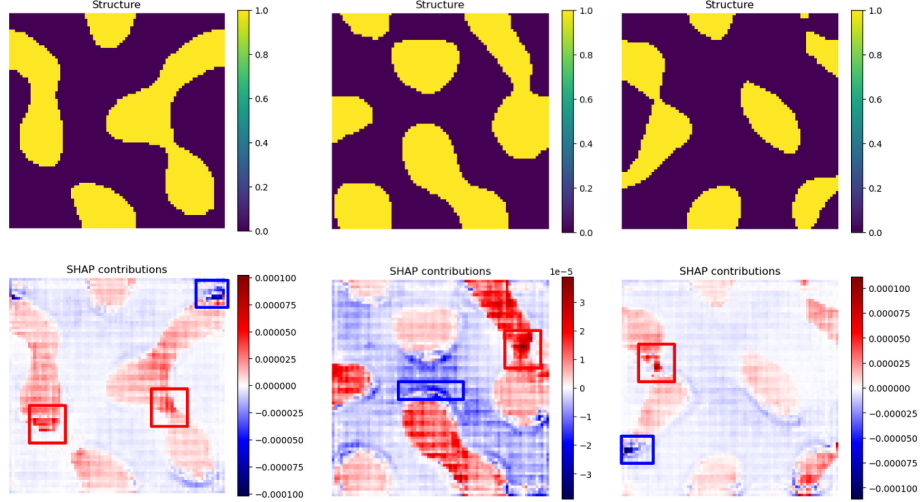


Fig. S4: Intersection planes through 3D structure and corresponding spatial distribution of SHAP values. Red (blue) color indicates regions with positive (negative) impact on the predicted sample stiffness.

shown in red, denote areas that contribute positively to the overall stiffness of the structure. Several of the most influential regions, highlighted with red boxes, coincide with material-rich areas, particularly narrow struts or thin connections within the network. The high SHAP values in these regions indicate that their removal (i.e., replacement with void) would result in the largest reduction in stiffness, emphasizing their critical role in sustaining the mechanical response of the sample. Put differently, red regions identify the structural motifs with the strongest positive contribution to stiffness.

In contrast, regions with negative SHAP values, shown in blue, are typically located adjacent to pores or vacuum regions. Dark blue zones, highlighted with blue boxes, where the SHAP values are large in magnitude but negative, are most often located near the structure boundaries. Dark blue zones, highlighted with blue boxes, are most often found near the structure boundaries. These negative attributions imply that the presence of a void in these areas has a large negative impact on mechanical strength. The model suggests that adding material here could yield the greatest relative improvement in stiffness. In other words, while these empty voxels currently contribute negatively to the mechanical load-bearing capacity, they represent the regions with the highest potential for improvement: if solid material were introduced here, the structure could achieve a significant enhancement in its overall stiffness.

This interpretation highlights the dual role of SHAP analysis: it not only identifies the structural features most critical for maintaining stiffness but also pinpoints the “weak links” where targeted modifications could most effectively reinforce mechanical performance. Unlike raw molecular dynamics simulations, which primarily provide

numerical estimates of properties, the deep learning model, together with SHAP analysis, enables a systematic approach to improving the mechanical strength of a given sample.

Besides SHAP analysis, we visualized how the input structure’s representation transforms inside the model. Figure S5 shows 2D slices of a voxelized input as it passes through initial convolution and DenseBlocks. The initial convolutional filters (3 out of 64 shown) activate on low-level features: one filter responds primarily to solid regions of a structure, another activates on voids, and a third responds strongly at narrow necks where ligaments connect. After DenseBlock1, some of the activation maps still reflect the initial input geometry, as the network operates on high-resolution data at this shallow depth. Deeper layers of the network become progressively more abstract. The activation volumes observed after DenseBlock2, DenseBlock3, and DenseBlock4 have sizes of 10×10 , 5×5 , and 2×2 , respectively. These low-resolution feature maps do not reflect voxelized nanoporous structure but instead encode higher-level, mesoscale, and global structural features that are relevant for predicting elastic constants.

4 Generalization test on architected structures

We tested the best-performing model (trained only on stochastic nanoporous structures) on a separate set of 22 topologies from the RCSR (Reticular Chemistry Structure Resource) database. Each of the topologies was generated with five values of target solid fraction (0.1, 0.2, 0.3, 0.4, 0.5 - recalculated later for as-built solid fraction), resulting in a total of 110 structures; examples are visualized in Figure S6. This set was labeled with MD-computed elastic constants, following the same procedure as the stochastic samples.

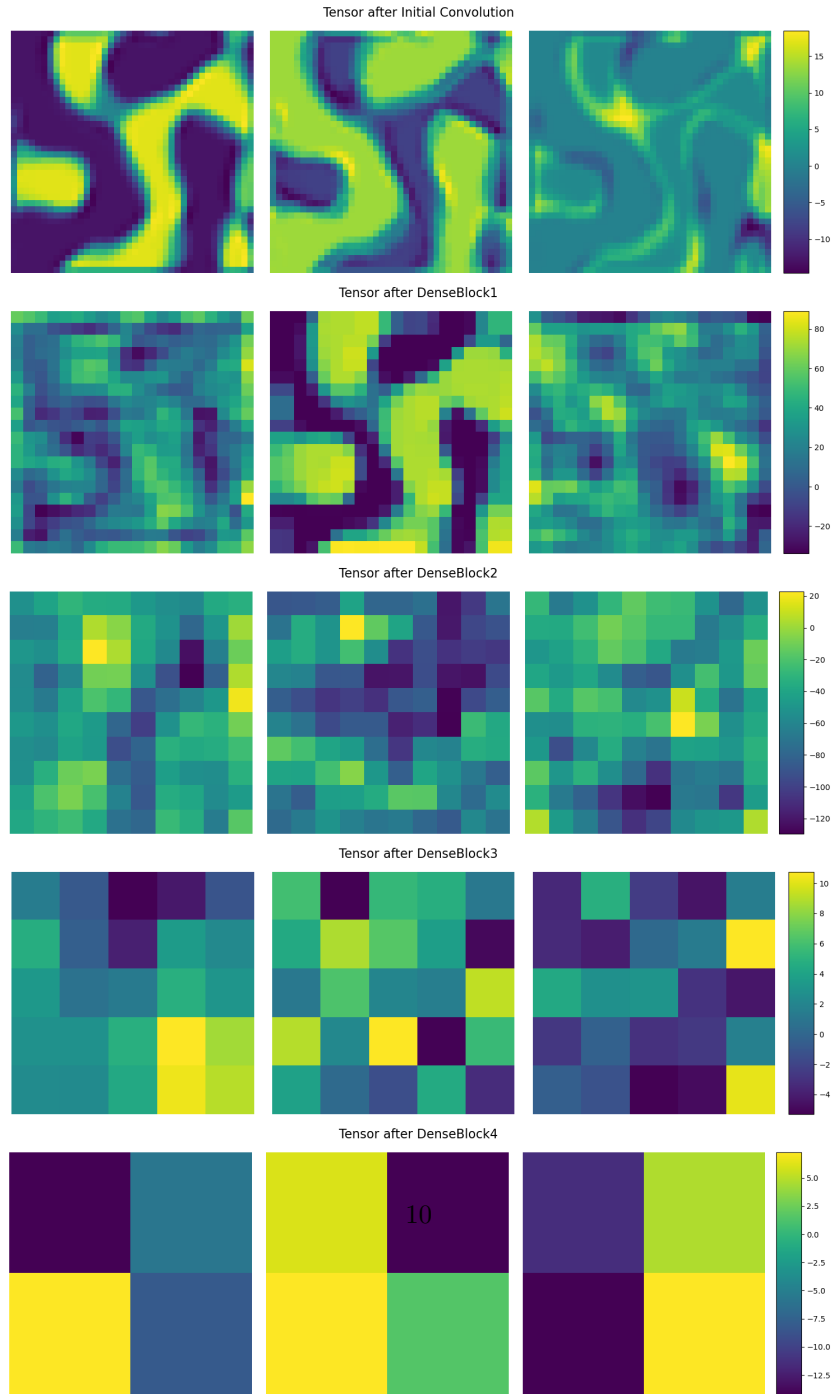
Following the same procedure as with stochastic structures, topologically defined samples were converted to a binary array of $80\times 80\times 80$ voxels to obtain CNN-predicted values of elastic constants. When the model was directly applied to architected structures with no fine-tuning, prediction errors increased substantially, compared to the model’s performance on the stochastic test set. This behavior is expected, as the model was never trained on architected structures and cannot accurately extrapolate to unseen data. Despite the large prediction errors, the model preserved the relative ordering of stiffness across the new dataset. The Spearman rank correlation between MD-calculated and CNN-predicted elastic constants is $\rho = 0.955$ and p-value = 3.26×10^{-58} , indicating a strong monotonic agreement. In practice, it means that the model is able to identify which topologies have higher or lower elasticity compared to others, even though its absolute values of predictions are biased.

5 Hybrid model architecture

This architecture is a hybrid neural network model that fuses convolutional and dense feature representations to enable joint learning from binary vectorized structural inputs and handcrafted descriptor-based features. It is designed to perform a regression task where both spatial and tabular data provide complementary information.



(a) Input tensor slice (binary).



(b) Intermediate-layer activations (Initial Conv + DenseBlocks).

Fig. S5: Visualization of DenseNet-201 intermediate layers.

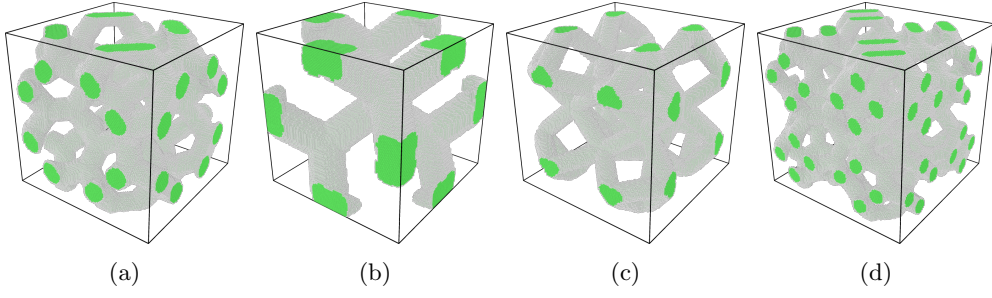


Fig. S6: Examples of topologically defined structures.

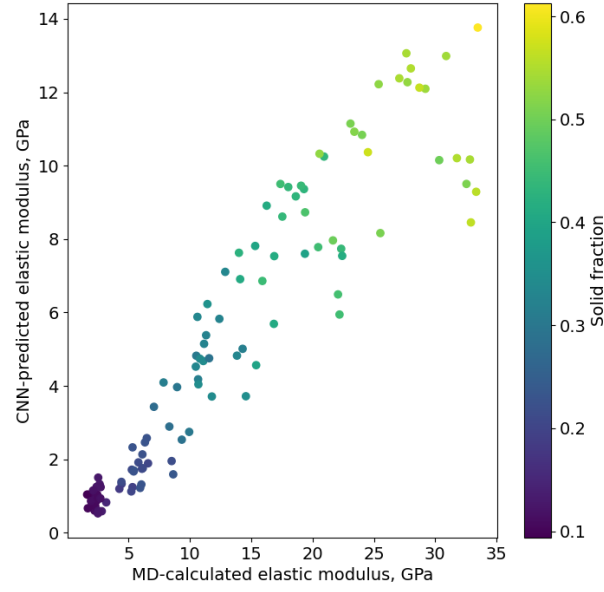


Fig. S7: MD-calculated vs CNN-predicted elastic constants for dataset of RCSR topologies.

The first branch uses a convolutional neural network to extract high-level features from structured input data. This part of the model processes the data through a convolutional backbone (such as DenseNet-201) to produce a feature vector (denoted in red in the Figure S8). These extracted features represent complex spatial or hierarchical patterns present in the input. In the case of DenseNet-201, this feature vector has a dimension of 768.

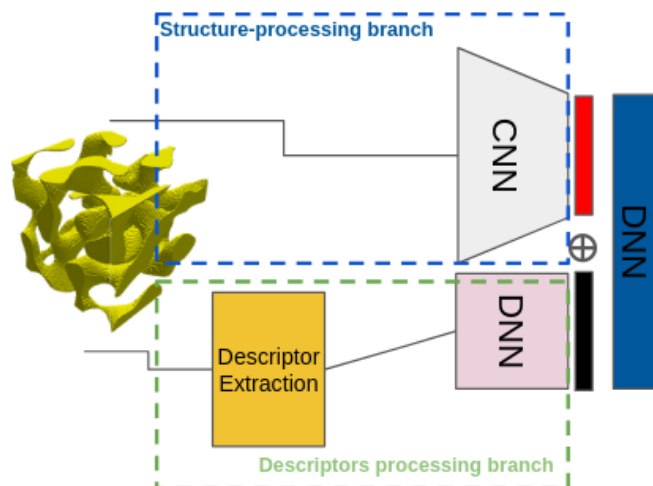


Fig. S8: Hybrid model architecture.

The second branch handles a separate input consisting of numerical descriptors as listed in Table 1 of the main manuscript. This input is passed through a sequence of linear transformations, activation functions, and dropout layers (depicted by pink rectangle in the Figure S8) designed to distill the information into a compact, informative representation. This fully-connected network consists of 3 linear layers, each having 33 nodes, connected with ReLU activation function and drop-out layers with dropout probability of 0.3 and produces a feature vector (denoted as black rectangle) of dimension 33. Once both branches have processed their respective inputs, their outputs — one from the CNN and one from the fully connected network — are concatenated into a single combined feature vector. This combined representation is then passed through another sequence of fully connected layers (represented as blue DNN network) that further integrate the information and transform it down to a final prediction, typically a single output value.

References

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